

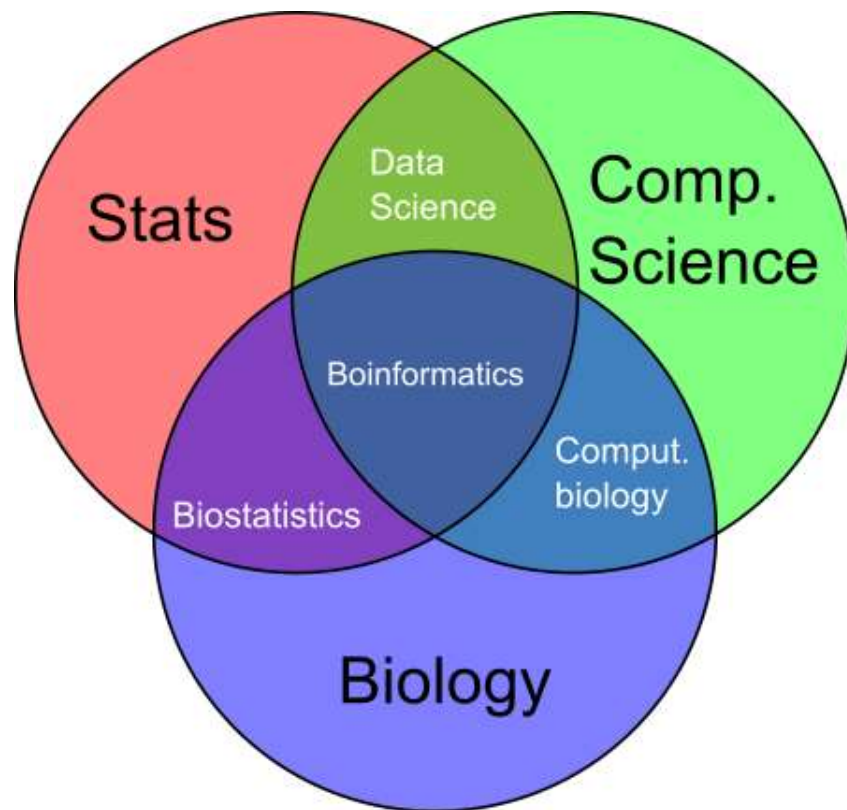
Bioinformatics – An Introduction

Hassaan Mehboob Awan, Ph.D.
Assistant Professor
Department of Biosciences
COMSATS University Islamabad (CUI)

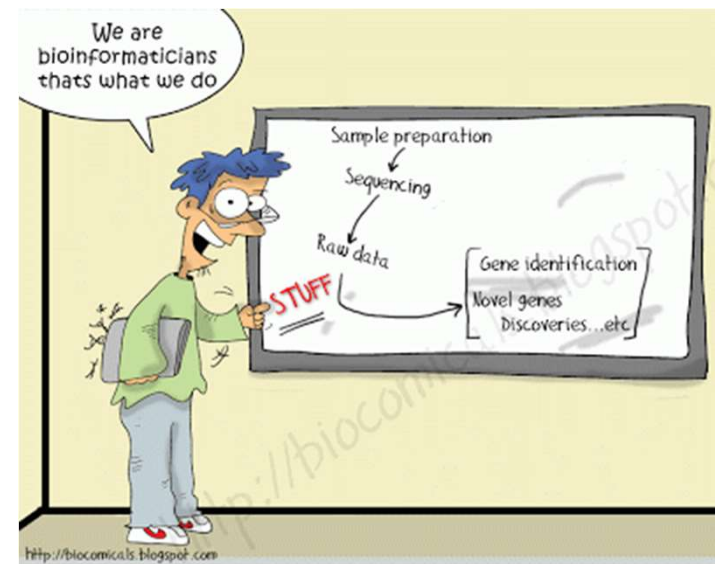


- Bioinformatics is an interdisciplinary field that develop methods and software tools for understanding biological data.
- Union of biology and informatics: bioinformatics involves the technology that uses computers for storage, retrieval, manipulation, and distribution of information related to biological macromolecules such as DNA, RNA, & proteins.
- Bioinformatics can be considered as a field of data science for solving problems in biology and medicine.
- As an interdisciplinary field of science, bioinformatics combines computer science, statistics, and mathematics to analyze and interpret biological data.





<https://genomejigsaw.files.wordpress.com>



<http://4.bp.blogspot.com/>





Paulien Hogeweg



Ben Hesper

Bioinformatics: a brief history

Early History: Bioinformatics, a Work Concept

In the beginning of the 1970s, Ben Hesper and I started to use the term “bioinformatics” for the research we wanted to do, defining it as “the study of informatic processes in biotic systems” (Although several public sources [see below] trace the origin of the term to publications by us that appeared in 1978 [1], [2], in fact we were using it as early as 1970, proposing the definition above in an article in Dutch that is not generally accessible [3].)

<https://doi.org/10.1371/journal.pcbi.1002021>



Bioinformatics: a brief history

Three important factors facilitated the emergence of computational biology during early 1960s.

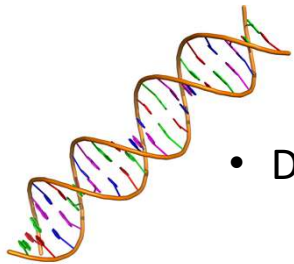
- Idea that macromolecules carry information
- Expanding collection of amino-acid sequences
- High speed digital computers



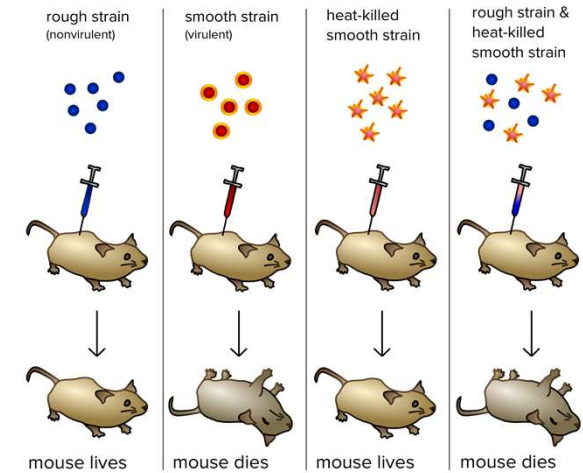
Bioinformatics: a brief history

- Avery – MacLeod – McCarty experiment (1944)

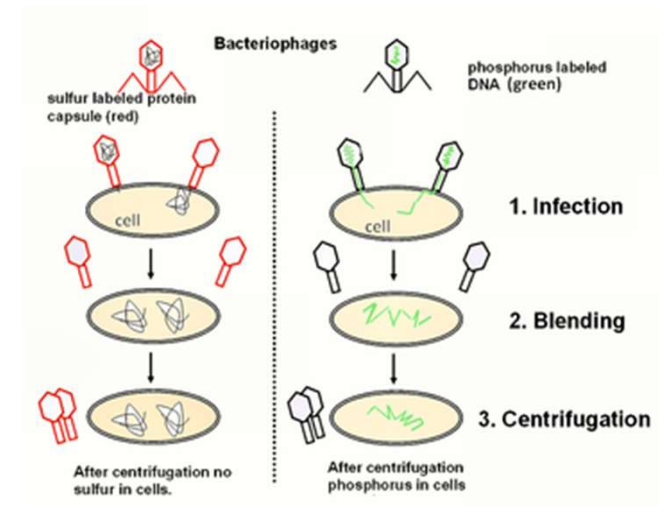
- Hershey & Chase experiment (1952)



- DNA double helical structure (1953)



Picture taken from: [Khan Academy](https://www.khanacademy.org/a/avery-macleod-mccarty-experiment-a-1944-experiment-that-proved-dna-is-the-genetic-material/a/avery-macleod-mccarty-experiment-a-1944-experiment-that-proved-dna-is-the-genetic-material/a/avery-macleod-mccarty-experiment-a-1944-experiment-that-proved-dna-is-the-genetic-material)



Bioinformatics: a brief history

Protein analysis was the starting point!

- First sequence of protein (Insulin) was published – 1950s
- Atlas of Protein Sequence and Structure – 1965
- Development of Protein Data Bank (PDB) – 1970s
- Sequence Alignment algorithm development – 1970s
- First protein structure prediction algorithm developed – 1974

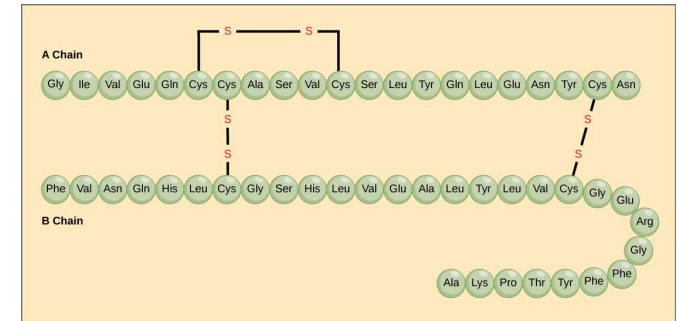


Figure 4 | The IBM 7090 computer, which Margaret Dayhoff used for her early work. This famous computer was one of the first solid-state machines and was used widely in business and defence settings, as well as scientific applications. (Photograph courtesy of IBM archives.)

Margret Dyhoff – the first bioinformatician

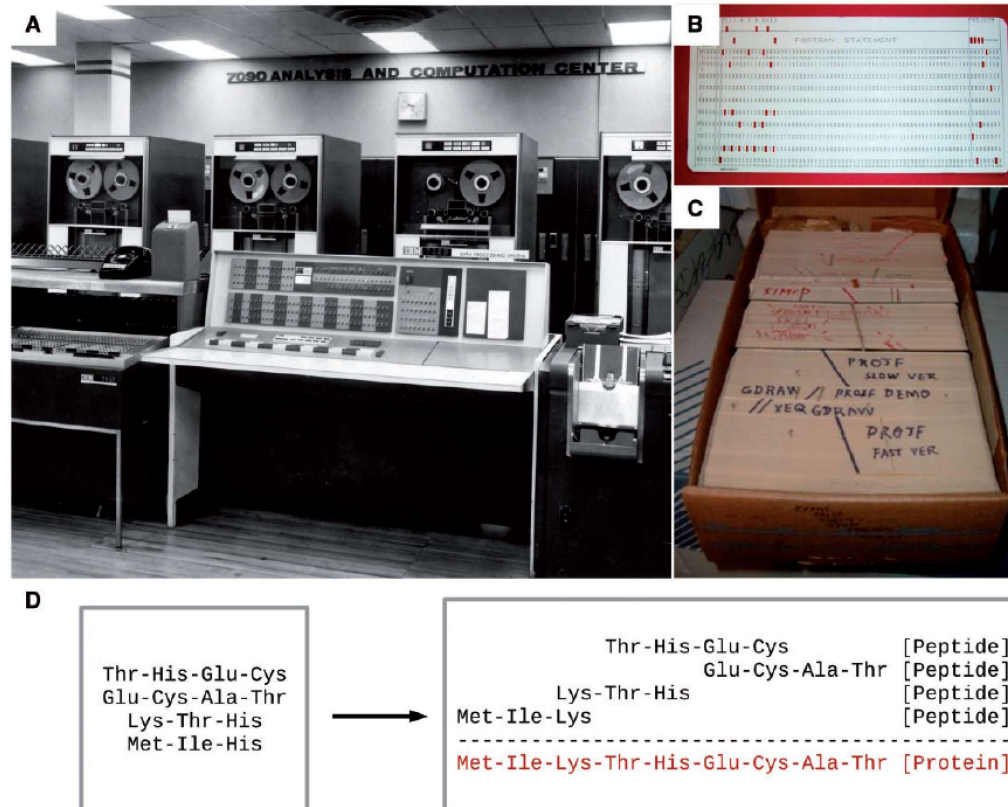
the mother and father of bioinformatics



Major contributions

1. Pioneer of application of computational methods in the field of biochemistry
2. Her collaboration with Robert S. Ledley (physicist) resulted in the development of COMPROTEIN, a computer program designed to determine the protein primary structure using Edman degradation method of peptide sequencing.
3. Developed the one-letter amino acid code that is still in use today.
4. Published the first *Atlas of Protein Sequence and Structure* in 1965





COMPROTEIN, the first bioinformatics software



We are in the age of *In Silico* biology!

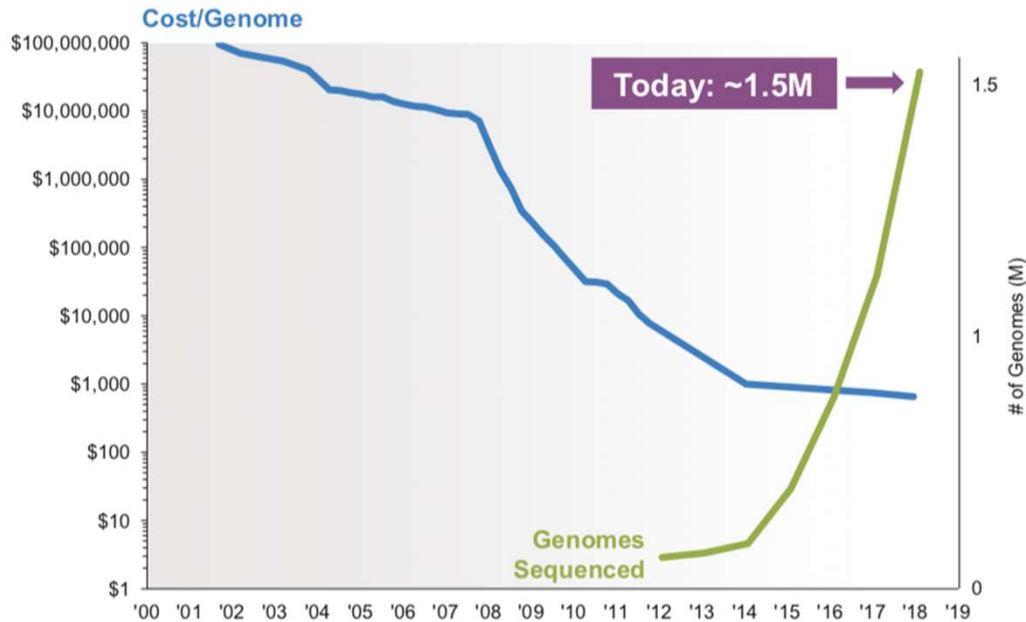


<http://www.anaxomics.com/>

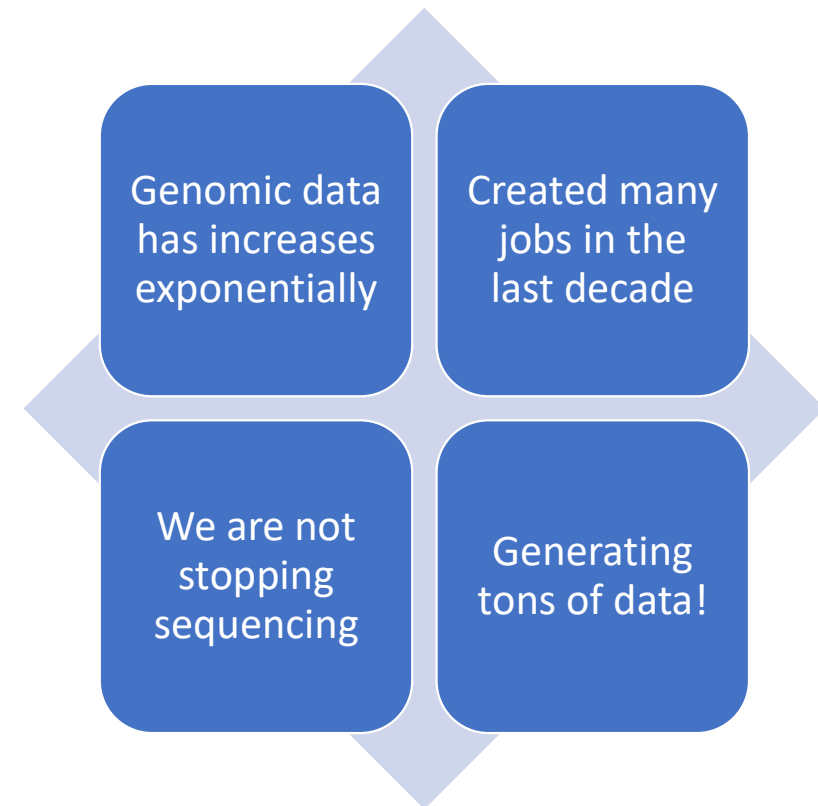
- Biology is becoming quantitative science
- Transformation from observational to experimental to data science
- New discoveries require large datasets and advanced analytical methods

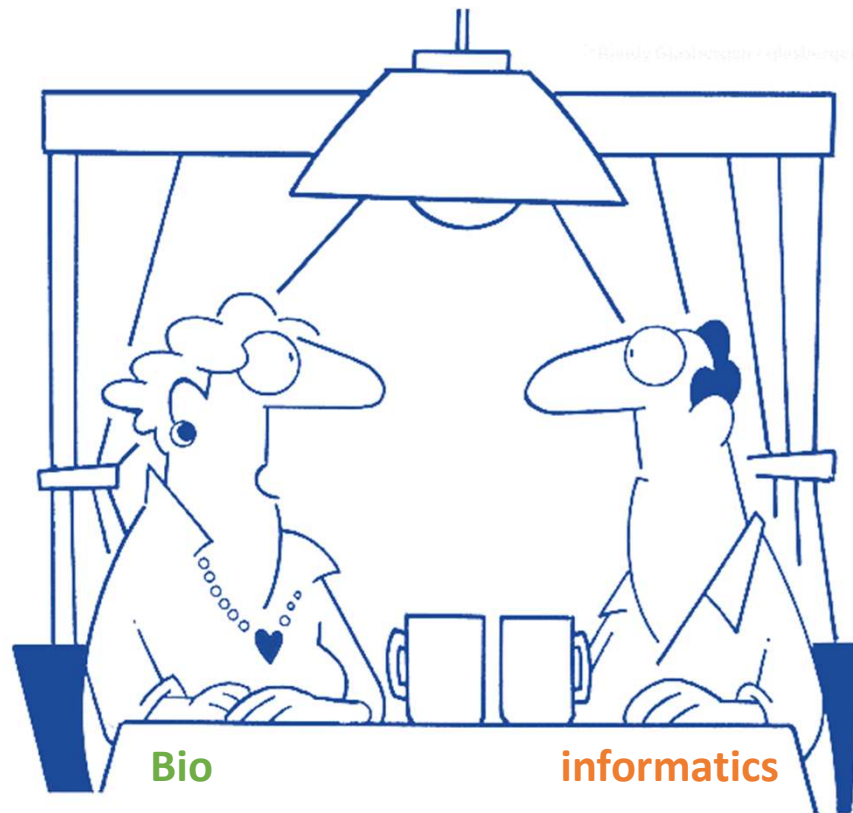


The power of genomics!



forbes.com





"Our love gets deeper and deeper every day."



Goals

The ultimate goal of bioinformatics is to better understand a living cell and how it functions at the molecular level



Scope

Development of computational tools and databases.

Application of these tools and databases.



Applications

Structure analysis

Nucleic acid
structure
prediction

Protein structure
prediction

Protein structure
classification

Protein structure
comparison

Sequence analysis

Genome
comparison

Phylogeny

Gene & promoter
predication

Motif discovery

Seq. database
search

Seq. alignment

Function analysis

Metabolic
pathway
modeling

Gene expression
profiling

Protein
interaction
prediction

Protein
subcellular
localization
prediction

Software development
Database construction and curation



Bioinformatics is cool!



<http://www.biocomics.com>

- Offers flexibility
- Can work from most places
- Don't need much to do science!



Nick Loman
@pathogenomenick

 Follow

Best thing about doing bioinformatics; still being able to work with the baby sleeping on you. Try that in the lab.

RETWEETS
29

LIKES
16



6:58 AM - 31 Jul 2012



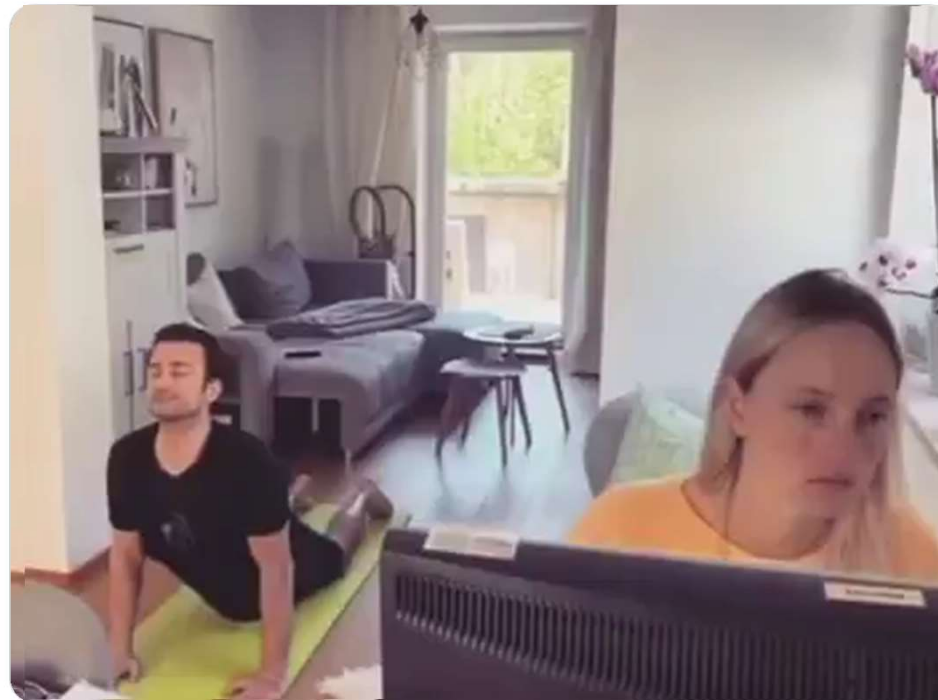


Maria Alejandra Sierra

@asierra93



When the wife works in bioinformatics and the husband in wet lab [#QuarentineDays](#)



9:13 PM · Apr 22, 2020 · Twitter for Android

958 Retweets **169** Quote Tweets **6.9K** Likes

